

Incident Postmortem Collection 2025

Root Cause Analysis, Impact Assessment, and Remediation Actions

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Document Version History

Ver.	Date	Author	Description
0.1	2024-01-15	J. van den Berg	Initial draft
0.5	2024-03-22	M. de Vries	Peer review incorporated
1.0	2024-06-10	P. Jansen	Approved by Architecture Board
1.5	2024-11-04	J. van den Berg	Updated for Q4 changes
2.0	2025-02-18	M. de Vries	Annual revision
2.1	2025-04-01	P. Jansen	Minor corrections

Postmortem: INC-2025-001 - Kubernetes Cluster Node Exhaustion

Root Cause

The Kubernetes cluster on App Server Cluster ran out of resources due to memory leak in Claims Platform.

Impact

All services on the cluster were affected: Risk Calculator, FDE, Customer Self-Service Portal.

Remediation

Memory limit enforcement + HPA tuning

Lessons Learned

The regulatory compliance dashboard provides real-time visibility into the status of 47 compliance controls across GDPR, PCI-DSS, SOX, and Solvency II requirements.

The data governance framework establishes clear ownership, quality standards, and lifecycle policies for all enterprise data assets, ensuring compliance with GDPR, PCI-DSS, and industry-specific regulations.

Postmortem: INC-2025-002 - PostgreSQL Replication Lag

Root Cause

PostgreSQL 15 replication lag on Database Server Replica exceeded 30 seconds during peak claims processing.

Impact

Read queries from Policy Admin returned stale data.

Remediation

Increased replication slots + connection pooling

Lessons Learned

Quarterly architecture reviews have become a cornerstone of the governance process, bringing together stakeholders from business, technology, and operations to assess alignment between strategic objectives and technical implementation.

The integration architecture team has standardized on event-driven patterns using Apache Kafka for asynchronous communication and REST APIs with OpenAPI specifications for synchronous request-response interactions.

Postmortem: INC-2025-003 - Apache Kafka Partition Rebalance

Root Cause

Kafka triggered unexpected partition rebalance, causing 5-minute message processing delay.

Impact

The Integration Bus stopped receiving events, affecting Billing System and Notification Service.

Remediation

Static partition assignment + consumer group tuning

Lessons Learned

The microservices migration strategy follows a strangler fig pattern, gradually replacing monolithic components with independently deployable services while maintaining backward compatibility.

The enterprise architecture practice at ArchiSurance has evolved significantly over the past decade, transitioning from a documentation-centric approach to an active governance model that integrates continuous feedback loops with development teams.

Postmortem: INC-2025-004 - API Gateway Certificate Expiry

Root Cause

TLS certificate on Gateway expired, blocking all external traffic.

Impact

Customer-facing services (MyArchiSurance Portal, Broker Portal) were inaccessible for 23 minutes.

Remediation

Automated cert renewal via cert-manager

Lessons Learned

The data governance framework establishes clear ownership, quality standards, and lifecycle policies for all enterprise data assets, ensuring compliance with GDPR, PCI-DSS, and industry-specific regulations.

The adoption of cloud-native technologies has fundamentally changed how the organization thinks about infrastructure provisioning, moving from months-long procurement cycles to self-service platforms that enable teams to deploy in minutes.

Postmortem: INC-2025-005 - Fraud Detection Model Drift

Root Cause

The Anti-Fraud Module ML model (fraud-scoring-model.onnx) showed 40% increase in false positives after data distribution shift.

Impact

Claims Assessment process backed up with manual reviews.

Remediation

Model retraining pipeline + data drift monitoring

Lessons Learned

Capacity planning models project a 40% increase in transaction volumes over the next 18 months, driven by the expansion into the commercial insurance market and the launch of digital self-service channels.

Cost optimization initiatives in the cloud infrastructure have yielded a 22% reduction in monthly compute costs through right-sizing, committed use discounts, and automated scaling policies.

Postmortem: INC-2025-006 - Oracle Database Connection Pool Exhaustion

Root Cause

Oracle DB max connections reached, blocking Billing and Oracle DWH queries.

Impact

Invoice generation (Broker Commission Settlement) delayed by 4 hours.

Remediation

Connection pool increase + query optimization

Lessons Learned

The integration architecture team has standardized on event-driven patterns using Apache Kafka for asynchronous communication and REST APIs with OpenAPI specifications for synchronous request-response interactions.

Cost optimization initiatives in the cloud infrastructure have yielded a 22% reduction in monthly compute costs through right-sizing, committed use discounts, and automated scaling policies.

Postmortem: INC-2025-007 - DMZ Firewall Rule Misconfiguration

Root Cause

A change to DMZ Network firewall rules accidentally blocked traffic from F5 Big-IP to the application subnet.

Impact

All external traffic was blocked. Full outage for 12 minutes.

Remediation

Change review process + automated firewall testing

Lessons Learned

Cost optimization initiatives in the cloud infrastructure have yielded a 22% reduction in monthly compute costs through right-sizing, committed use discounts, and automated scaling policies.

Network segmentation has been implemented following a zero-trust model, with micro-segmentation between service tiers and mandatory mutual TLS for all inter-service communication.

Postmortem: INC-2025-008 - Redis Cache Eviction Storm

Root Cause

the in-memory cache evicted cached sessions due to memory pressure, causing Customer Portal users to be logged out.

Impact

~2000 active sessions lost. Customer complaints increased 300%.

Remediation

Redis maxmemory-policy tuned + session persistence added

Lessons Learned

Performance benchmarking has revealed that the current claims processing pipeline handles peak loads of 15,000 claims per hour, with 99th percentile latency under 2 seconds for API responses.

API management has been centralized through the API Gateway, which now handles over 2 million requests per day with an average response time of 120 milliseconds.

Postmortem: INC-2025-009 - Elasticsearch Index Corruption

Root Cause

Elasticsearch primary shard became unassigned on Monitoring Server, losing 3 hours of log data.

Impact

Audit trail gap for compliance (SOX Audit Trail).

Remediation

Snapshot schedule + replica count increase

Lessons Learned

The enterprise architecture practice at ArchiSurance has evolved significantly over the past decade, transitioning from a documentation-centric approach to an active governance model that integrates continuous feedback loops with development teams.

The adoption of cloud-native technologies has fundamentally changed how the organization thinks about infrastructure provisioning, moving from months-long procurement cycles to self-service platforms that enable teams to deploy in minutes.

Postmortem: INC-2025-010 - Cloud VPN Tunnel Flapping

Root Cause

The Cloud VPN tunnel to on-premise data center experienced intermittent connectivity, affecting Oracle DB access.

Impact

The ERP migration batch job failed after 6 retries.

Remediation

Dedicated interconnect provisioned

Lessons Learned

The adoption of cloud-native technologies has fundamentally changed how the organization thinks about infrastructure provisioning, moving from months-long procurement cycles to self-service platforms that enable teams to deploy in minutes.

The container orchestration platform (Kubernetes) runs 150 pods across 3 node pools, with horizontal pod autoscaling configured for all production workloads.